

A Local-Slope ACUG Transfer Theorem for the Graph Embedding Question

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Source Question

The source paper is M. Jung, C. Petitjean, A. Prochazka, A. Quilis, *Affine Approximation in Finite Nagata Dimension and Applications to Lipschitz-free spaces*, arXiv:2606.11723. Question 1 on page 34 asks:

Let M be a compact metric space admitting a superreflexive ACUG structure. Does M admit a bi-Lipschitz embedding into a compact quasiconvex metric space that also carries a superreflexive ACUG structure?

The same paper constructs, for every metric space M and every $\varepsilon > 0$, a graph space $G_\varepsilon(M)$ into which M embeds isometrically and which is compact when M is compact and $(1 + \varepsilon)$ -quasiconvex. The issue is whether ACUG structure transfers to $G_\varepsilon(M)$.

Proof Intuition

An ACUG structure controls approximants along rectifiable curves already present in M . The graph $G_\varepsilon(M)$, however, adds many new intervals $[x, y]$ between points of M . On a short new edge, the derivative of any extension must at least see the secant quotient

$$\frac{|g(x) - g(y)|}{d(x, y)}.$$

Thus ordinary upper-gradient control on M is not enough by itself: it may say nothing about secants between nearby points not joined by curves inside M .

The following theorem proves that this is the only obstruction for the graph method. If the ACUG approximants also come with a linear continuous majorant for small secants, then one extends approximants over the new graph edges, uses the secant majorant on short edges, and uses an ordinary one-dimensional derivative on long edges. The resulting majorant is continuous on the graph and remains superreflexive because only finite Hilbert sums of the original superreflexive model are added.

Definition 1 (local-slope ACUG). *Let M be compact and X a Banach space. We say that M has a local-slope ACUG X -structure if there is $C_0 \geq 1$ such that for every $\eta > 0$ there are a linear subspace $H_\eta \subset \text{Lip}(M)$ and a linear map*

$$\Lambda_\eta : H_\eta \longrightarrow C(M, X^*)$$

with the following properties.

(i) For every $f \in \text{Lip}(M)$ with $\text{Lip}(f) \leq 1$, there is $g \in H_\eta$ such that

$$\text{Lip}(g) \leq C_0, \quad \|f - g\|_\infty \leq \eta, \quad \|\Lambda_\eta g\|_\infty \leq C_0.$$

(ii) For every $g \in H_\eta$, the function $x \mapsto \|\Lambda_\eta g(x)\|$ is an upper gradient of g on M .

(iii) For every $g \in H_\eta$ and all $x, y \in M$ with $0 < d(x, y) \leq \eta$,

$$\frac{|g(x) - g(y)|}{d(x, y)} \leq \max\{\|\Lambda_\eta g(x)\|, \|\Lambda_\eta g(y)\|\}.$$

Theorem 1. *Let M be a compact metric space with a local-slope ACUG X -structure, where X is superreflexive. Then for every $\varepsilon > 0$, the graph space $G_\varepsilon(M)$ of Jung–Petitjean–Prochazka–Quilis admits an ACUG structure modeled on a superreflexive Banach space. Consequently M embeds isometrically into a compact $(1 + \varepsilon)$ -quasiconvex metric space carrying a superreflexive ACUG structure.*

Proof. Fix $\varepsilon > 0$, and let $G = G_\varepsilon(M)$. By Proposition 5.1 of the source paper, G is compact, $(1 + \varepsilon)$ -quasiconvex, and contains an isometric copy of M .

It remains to prove that G has an ACUG structure. Let $\alpha > 0$ be an approximation tolerance. Choose $\eta > 0$ so small that $(2C_0 + 4)\eta < \alpha$, and take $(H, \Lambda) = (H_\eta, \Lambda_\eta)$ from the local-slope ACUG structure of M . We construct a subspace $\tilde{H} \subset \text{Lip}(G)$ and a linear operator

$$\Psi : \tilde{H} \longrightarrow C(G, Y^*) \quad \text{where} \quad Y^* = (X^* \oplus X^* \oplus \mathbb{R})_2.$$

The predual model $Y = (X \oplus X \oplus \mathbb{R})_2$ is superreflexive because finite Hilbert sums preserve superreflexivity.

The elements of $\tilde{H}$ are functions $u \in \text{Lip}(G)$ satisfying the following linear conditions. First $u|_M \in H$. On each edge $[x, y] \subset G \setminus M$ with $d(x, y) \leq \eta$, u is the fixed linear smoothing of the endpoint data $u(x), u(y)$: it is constant in small endpoint collars, is C^1 on the edge interior, and

$$|u'(p)| \leq 4 \frac{|u(x) - u(y)|}{d(x, y)} \quad (p \in (x, y)).$$

The smoothing is chosen once and for all. Its derivative is supported in the middle half of the edge. On each edge with $d(x, y) > \eta$, u is C^1 on the edge interior and is constant on endpoint collars of length $\eta/4$. These requirements define a linear subspace.

For $h = u|_M$, define Ψu as follows. On M , set

$$\Psi u(x) = (4\Lambda h(x), 4\Lambda h(x), 0).$$

On a short edge $[x, y]$, set the third coordinate to 0. The first two coordinates are a fixed continuous interpolation which equals $(4\Lambda h(x), 4\Lambda h(x))$ near x , equals $(4\Lambda h(x), 4\Lambda h(y))$ on the middle half of the edge, and equals $(4\Lambda h(y), 4\Lambda h(y))$ near y . On a long edge, use the same endpoint interpolation in the first two coordinates and put the actual one-dimensional derivative u' in the third coordinate. The endpoint collars make $u' \rightarrow 0$ at vertices. Since Λh is continuous on compact M , the resulting Ψu is continuous on G . The definition is linear in u .

We next check upper-gradient domination. On portions of curves lying in M , the norm of Ψu dominates $\|\Lambda h\|$, hence dominates variation by the local-slope ACUG hypothesis. On a long edge,

the third coordinate is u' , so the norm of Ψu dominates the one-dimensional metric derivative of u . On a short edge, u' is supported where

$$\Psi u = (4\Lambda h(x), 4\Lambda h(y), 0),$$

and therefore

$$\|\Psi u\| \geq 4 \max\{\|\Lambda h(x)\|, \|\Lambda h(y)\|\} \geq 4 \frac{|h(x) - h(y)|}{d(x, y)} \geq |u'|.$$

Thus the pointwise local Lipschitz constant of u on G is bounded by $\|\Psi u\|$. By the standard local-Lipschitz-constant upper-gradient lemma used in the source paper, $\|\Psi u\|$ is an upper gradient of u .

It remains to verify approximation. Let $F \in \text{Lip}(G)$ with $\text{Lip}(F) \leq 1$. Choose $h \in H$ such that $\text{Lip}(h) \leq C_0$, $\|F|_M - h\|_\infty \leq \eta$, and $\|\Lambda h\|_\infty \leq C_0$. Extend h to a function $\tilde{h} \in \tilde{H}$ as follows.

On each short edge $[x, y]$ with $d(x, y) \leq \eta$, use the fixed smoothing of the endpoint values $h(x), h(y)$. Then every point of the edge is within $\eta/2$ of an endpoint, and the preceding derivative bound gives

$$|\tilde{h}(p) - F(p)| \leq (2C_0 + 2)\eta.$$

On each long edge, keep $\tilde{h} = h(x)$ and $\tilde{h} = h(y)$ on the endpoint collars, choose a partition of the middle with mesh between $\eta/4$ and $\eta/2$, prescribe the values of F at the partition points, and smooth linearly between consecutive values. Exactly as in the proof of Proposition 5.1's purely unrectifiable case in the source paper, this gives

$$\|\tilde{h} - F\|_\infty \leq 4\eta, \quad \text{Lip}(\tilde{h}) \leq 8 + 2C_0.$$

Increasing the structural constant once, independent of α , we have

$$\text{Lip}(\tilde{h}) \leq C', \quad \|\Psi \tilde{h}\|_\infty \leq C', \quad \|\tilde{h} - F\|_\infty \leq \alpha.$$

This is precisely an ACUG Y -structure on G . □

Verification and Scope

The theorem is a partial answer to the source question. It proves the graph embedding conclusion under a stronger hypothesis than ordinary ACUG. The strengthening is not cosmetic: it controls the small secants that become derivatives on the new edges of $G_\varepsilon(M)$.

The full source question would follow if every compact superreflexive ACUG space admitted this local-slope upgrade, perhaps after changing the approximating subspaces. No proof of that upgrade is claimed here.

Novelty Check

Local repository indexes contained only one previous attempt for arXiv:2606.11723, which triaged the broad ACUG embedding question and did not promote a result. Web searches on June 20, 2026 for exact ACUG embedding phrases found the source paper arXiv:2606.11723 and the background paper Aliaga–Pernecka–Quero arXiv:2411.15107, but no later answer to Question 1.

References

- M. Jung, C. Petitjean, A. Prochazka, A. Quilis, *Affine Approximation in Finite Nagata Dimension and Applications to Lipschitz-free spaces*, arXiv:2606.11723.
- R. J. Aliaga, E. Pernecka, A. Quero, *Pelczynski's property (V^*) in Lipschitz-free spaces*, arXiv:2411.15107.